Physics Worksheet

Kinematics

Name: ____________________ **Student #:** _______________ **Student Email #:** ________

1. The acceleration of a cricket jumping sideways is observed to increase linearly with time, that is, $ax(t) = a_0 + jt$, where a_0 and j are constants. What can you say about the velocity of the cricket as a function of time?

   (a) It is constant

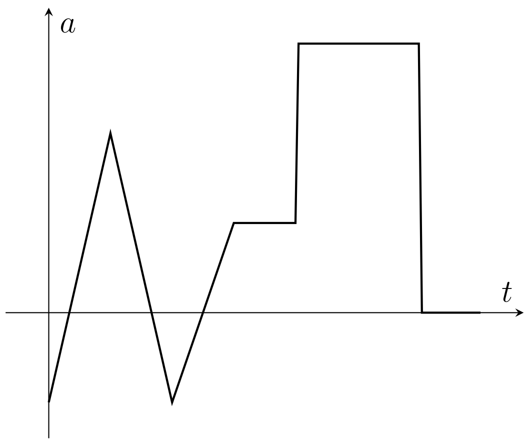
   (b) It increases linearly with time ($v(t) \propto t$)

   (c) It increases quadratically with time ($v(t) \propto t^2$)

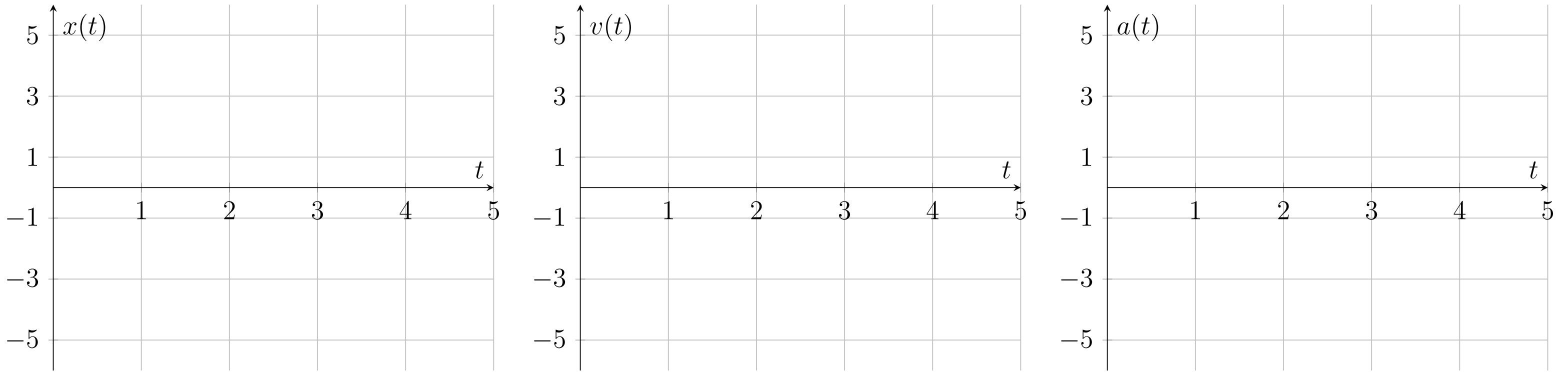
   (d) It increases with the cube of time ($v(t) \propto t^3$)

   (c)

2. Plot below the $v(t)$ and $x(t)$ for the given $a(t)$ plot. Assume $v_0 = 0$ and $x_0 = 0$



3. Below, plot the function $x(t) = 2x^{\frac{1}{2}}$, plot this as well as $v(t)$ and $a(t)$ of the same function



4. A large motor is running at 4000 rpm with an internal linear velocity of $v = 20\,\mathrm{m\,s^{-1}}$. What is the radius R of the motor and the frequency f at which the motor is rotating?

5. Enzo is driving a remote control car on a train that has just begun to move after making a stop at a station

(a) If the train is accelerating at $a = 2\,\mathrm{m\,s^{-2}}$ at $t = 0s$ and Enzo's toy car is going at a constant speed of $v = -3\,\mathrm{m\,s^{-1}}$, what is the toy car's position as a function of time as seen from the stationary platform of the train station?

(b) At what point in time will the toy car appear to be moving in the same direction as the train?